

High-level intuition & course overview



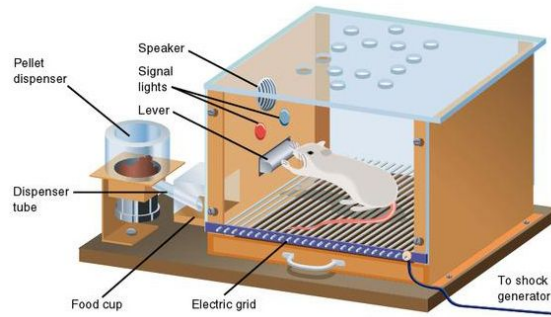
Introduction to Reinforcement Learning

Leiden University, The Netherlands

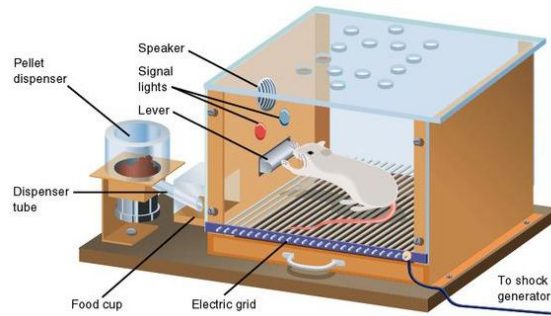
Biology



Biology



Biology

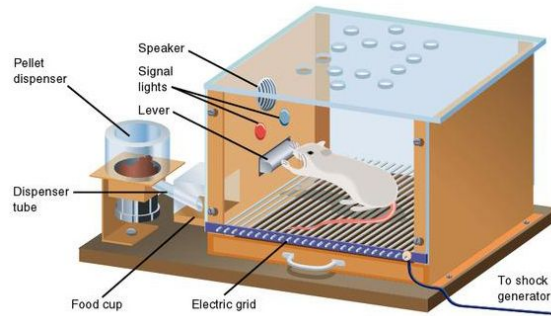


Skinner box



B.F. Skinner (1904 – 1990)

Biology



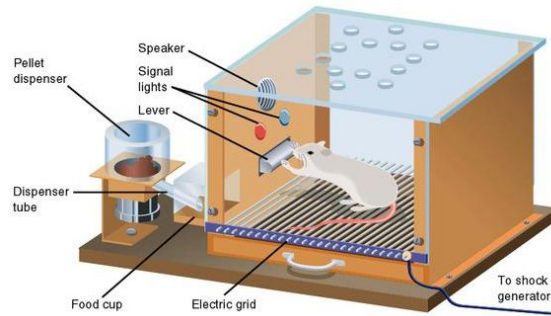
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Instrumental conditioning:
Learning behaviour based on reward and punishment (trial and error)

Biology



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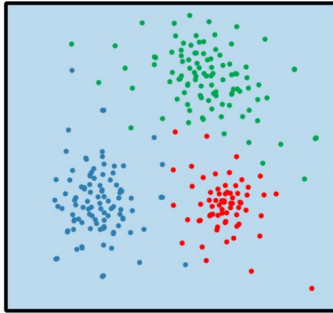
RL is the computational specification of this idea

machine learning

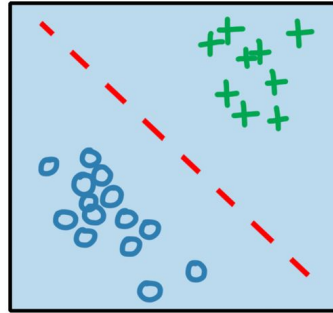


machine learning

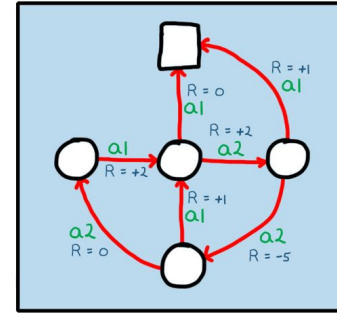
unsupervised
learning



supervised
learning



reinforcement
learning



Supervised vs reinforcement learning



Supervised vs reinforcement learning

Supervised learning

Reinforcement learning

Dataset

Feedback

Supervised vs reinforcement learning

Supervised learning

Reinforcement learning

Dataset

Given

Feedback

Supervised vs reinforcement learning

Supervised learning

Reinforcement learning

Dataset

Given

Active collection

Feedback

Supervised vs reinforcement learning

	Supervised learning	Reinforcement learning
<u>Dataset</u>	Given	Active collection
<u>Feedback</u>	Full (x with correct y)	

Supervised vs reinforcement learning

	Supervised learning	Reinforcement learning
<u>Dataset</u>	Given	Active collection
<u>Feedback</u>	Full (x with correct y)	Partial (state with correct action) (feedback on some outcomes)

Benefits of Reinforcement Learning



Benefits of Reinforcement Learning

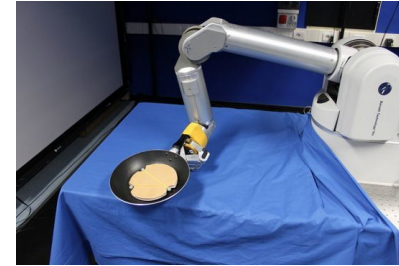


Autonomous behaviour/learning
(only specify goals)

Benefits of Reinforcement Learning



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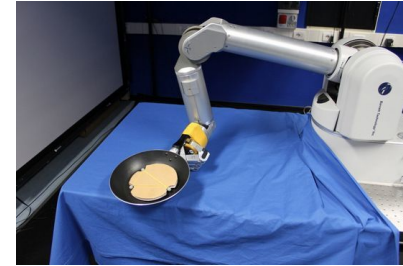


Solve tasks that you can't label
(only need to label the outcome)

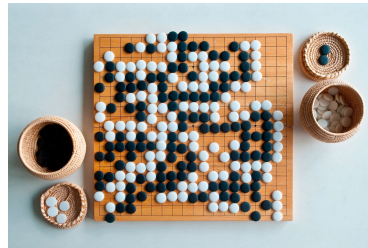
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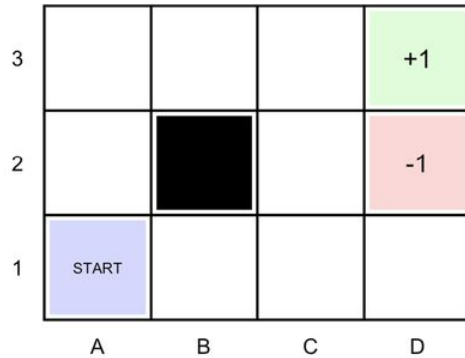


Outperform human solution
(only need to label the outcome)

Sequential decision-making

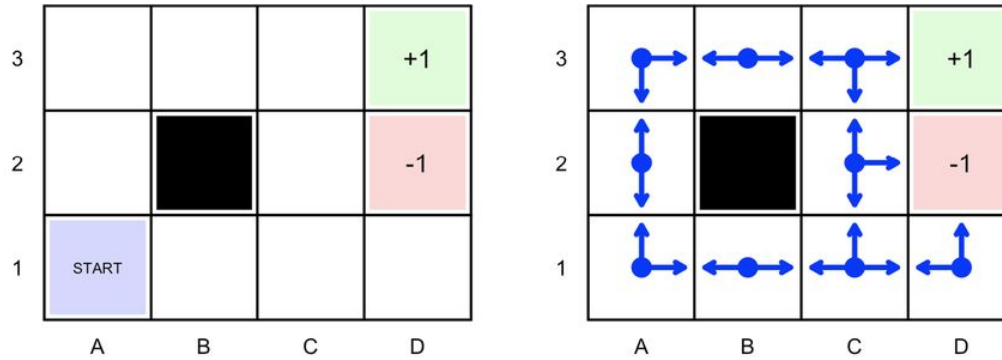


Sequential decision-making



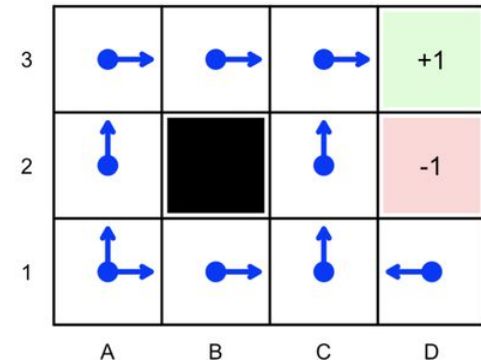
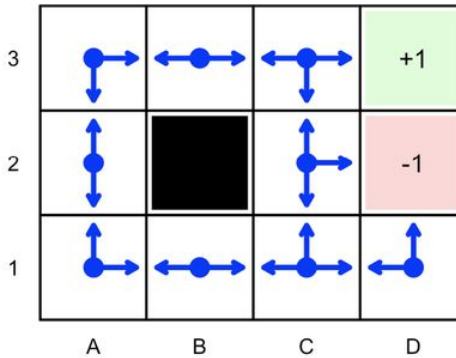
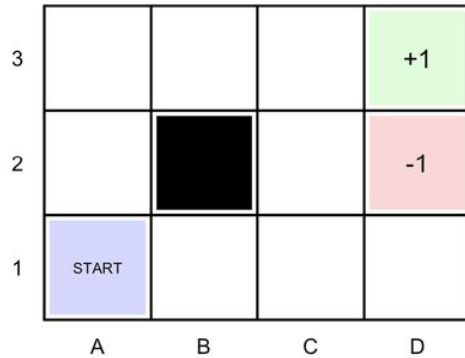
We start in a certain state, we can take actions that may bring us to other states, and there are some unknown rewards (goals) and punishments

Sequential decision-making



In advance we have no clue how to act in every state (*random policy*)

Sequential decision-making



We eventually want to find the action selection strategy in each state that in the long run gets us as much total reward as possible (the *optimal policy*)

Course structure

Block I



Block II



Block III



Block I:

Exploration & Dynamic Programming

Week 2: Exploration

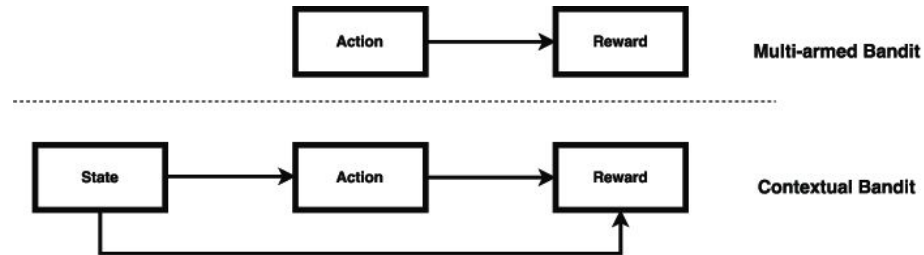


Week 2: Exploration

Start with the simplest one-step decision making problem: **Multi-armed Bandit**

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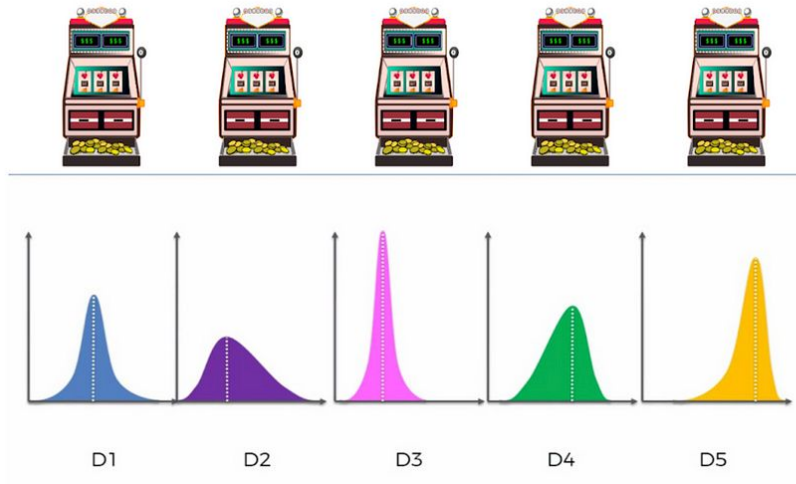


Week 2: Exploration

Key challenge: **exploration-exploitation trade-off**

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Multiple available actions with unknown reward distributions

Learn ways to trade-off exploration (something novel) & exploitation (known to work well)

Week 3: Markov Decision Process

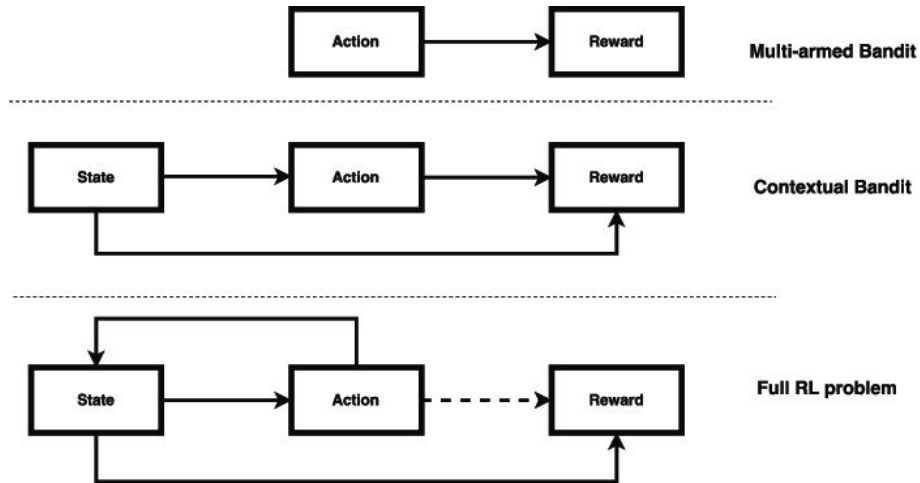


Week 3: Markov Decision Process

Make the decision-making problem *sequential*

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Week 3: Markov Decision Process



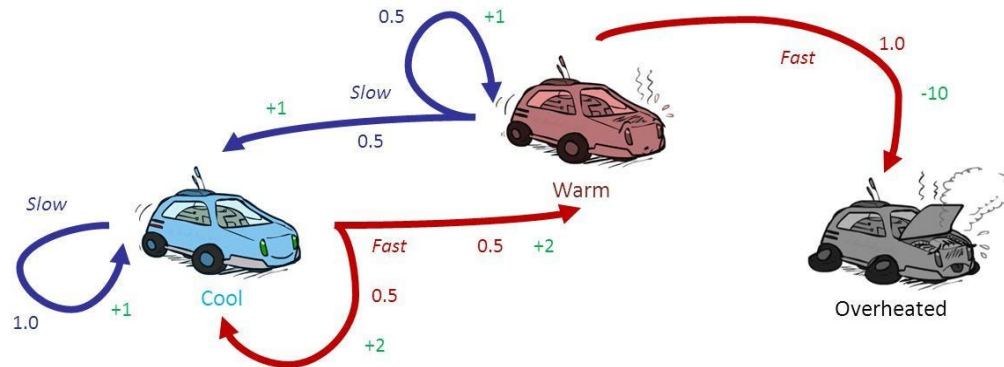
Week 3: Markov Decision Process

Learn to define sequential decision-making problems in a generic way



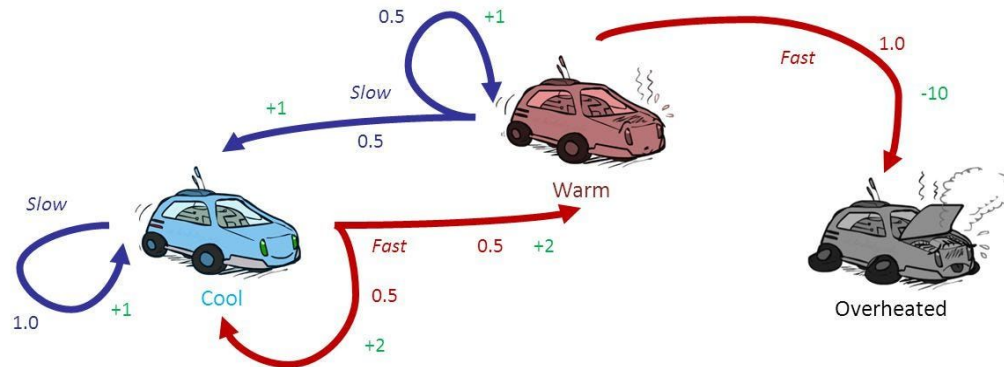
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Learn to define sequential decision-making problems in a generic way



- States
- Actions
- Transition probabilities
- Reward function

- Policy
- Value function

Week 4: Dynamic Programming



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If we have a *full transition model* we can *exactly* solve for the optimal policy

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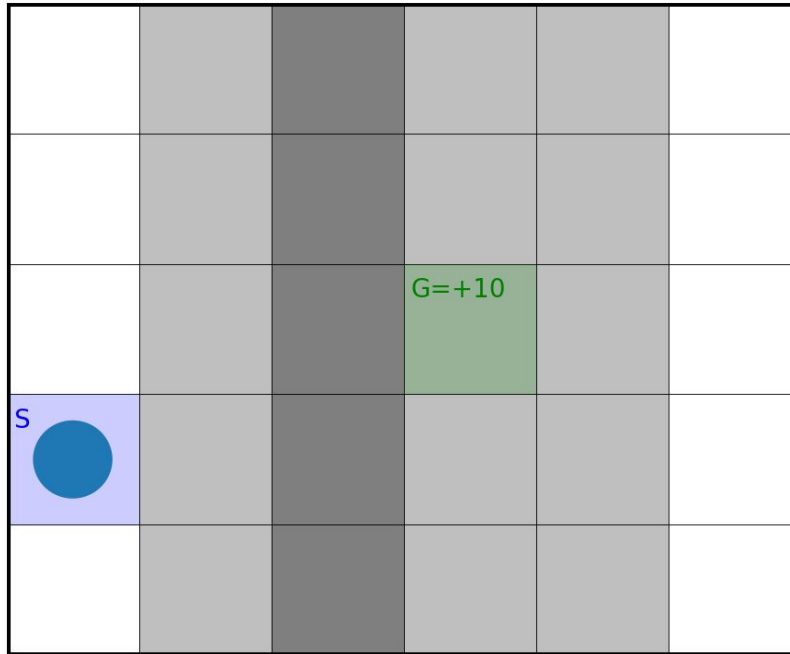
By:

- 1) Sweeping through all states, each time propagating rewards 1 step further back
- 2) Until we converge on the 'optimal value' (highest total reward obtainable) of each state-action

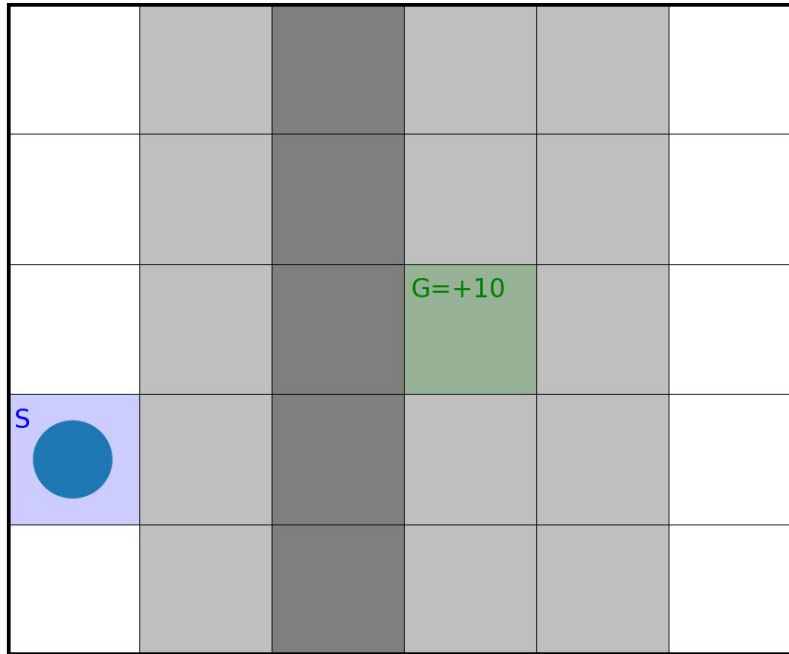
Week 4: Dynamic Programming - demonstration



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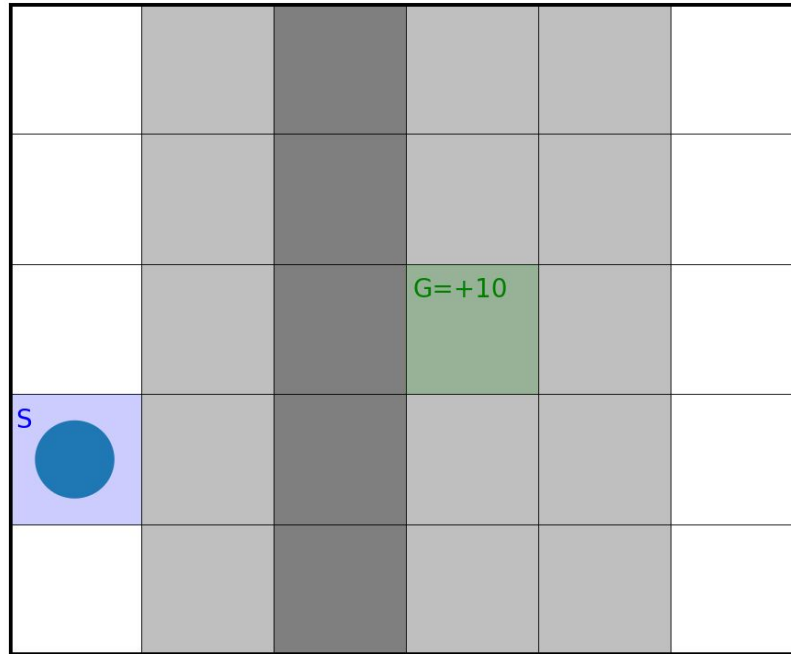


Week 4: Dynamic Programming - demonstration



- **States:** all locations
- **Actions:** Can move up/down/left/right
- **Transitions:** In the dark columns there blows an upward wind (up) with strength 1 (light grey) or 2 (dark grey)
- **Rewards:** The goal has a reward of +10, all other steps have a reward of -1

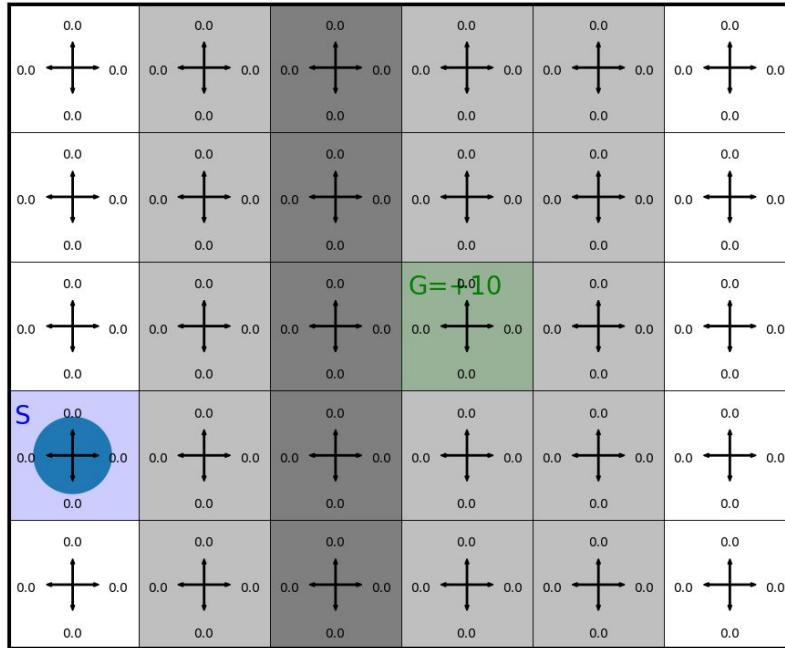
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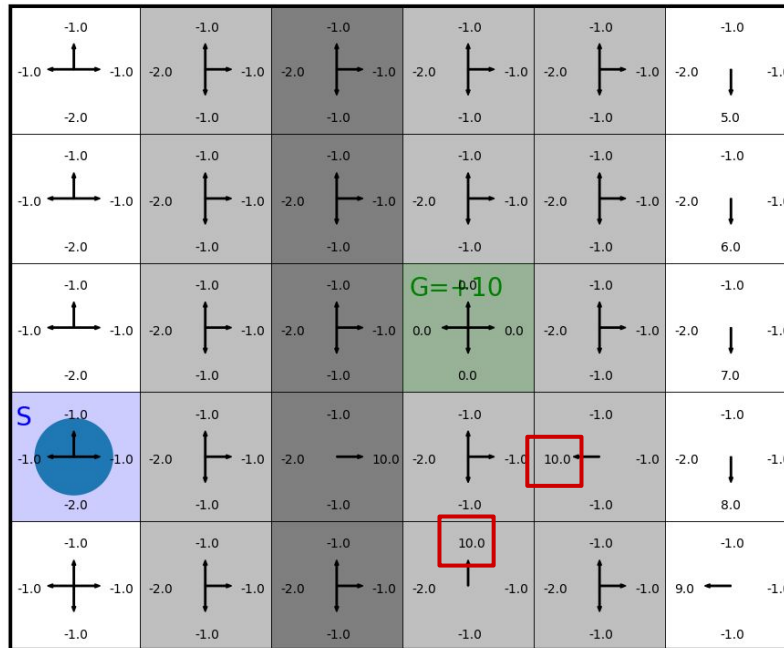
Question: What path do you think an optimal agent take?

Week 4: Dynamic Programming - demonstration



Initialize all state-action values
(e.g. to 0.0)

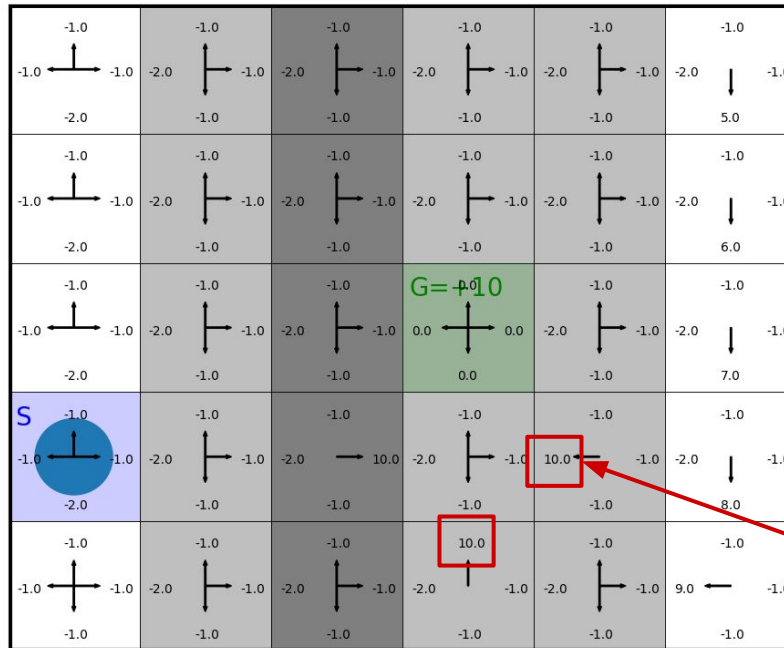
Week 4: Dynamic Programming - demonstration



Iterate (sweep through all states)

→ Rewards start propagating in value estimates

Week 4: Dynamic Programming - demonstration

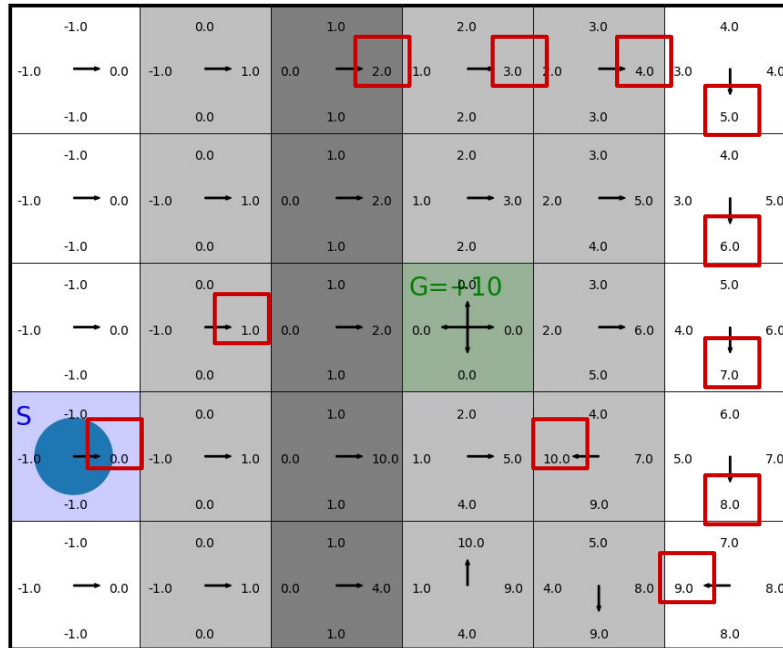


Iterate (sweep through all states)

→ Rewards start propagating in value estimates

'From this action, you can at best get a total reward of 10'

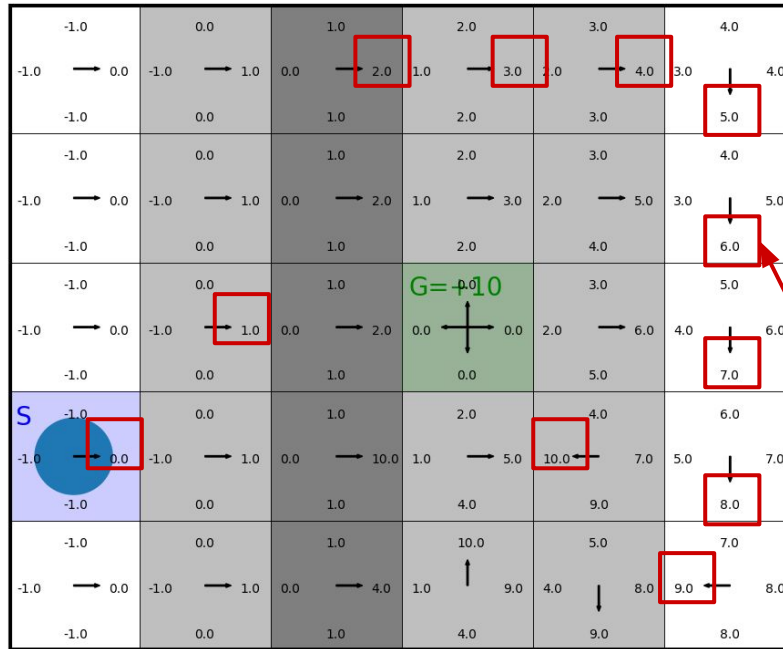
Week 4: Dynamic Programming - demonstration



Until convergence

→ All reward information has propagated

Week 4: Dynamic Programming - demonstration

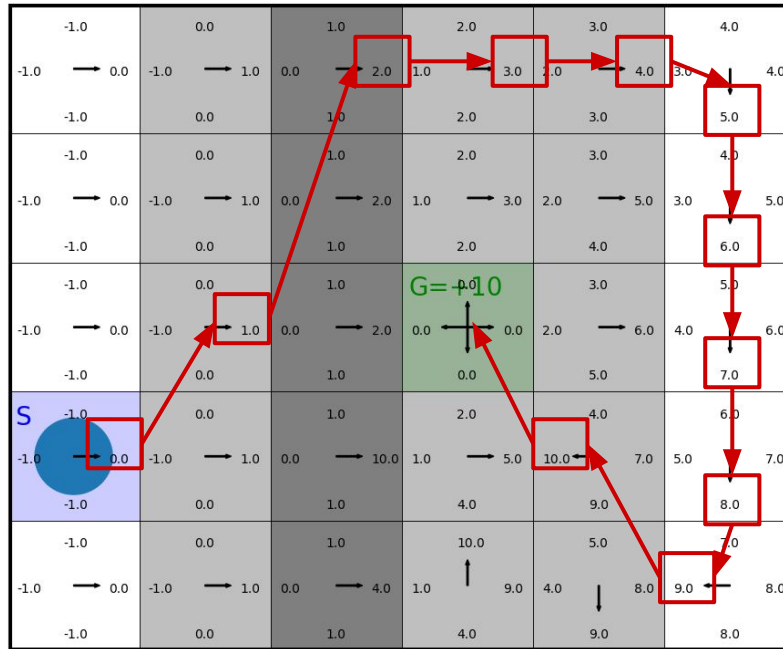


Until convergence

→ All reward information has propagated

*'From this action, you can at best get a total reward of 6'
(4 steps of -1, and one final reward of +10)*

Week 4: Dynamic Programming - demonstration



*Found the optimal policy:
by repeatedly taking the available
action with the highest value
estimate*

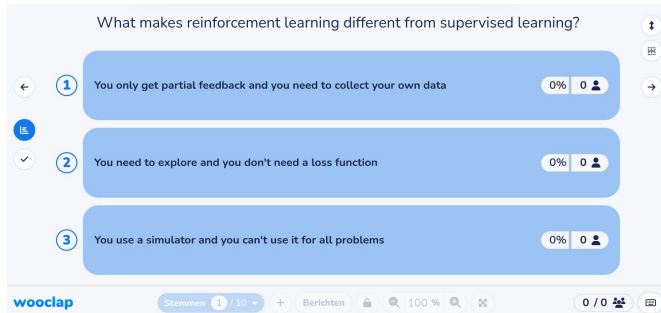
Week 5: Q&A + Quiz + Assignment 1 deadline

In the last week of each block you get time to reflect & complete the assignment



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Decision-making problems

One-step + model unknown

Exploration

Sequential + model known

Markov Decision Process
Dynamic Programming

Q&A + Quiz + A1

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Sequential + model known

Exploration

**Markov Decision Process
Dynamic Programming**

Sequential +
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Q&A + Quiz + A1

Block II:

Model-free
Reinforcement Learning

Week 6+7: Model-free reinforcement learning

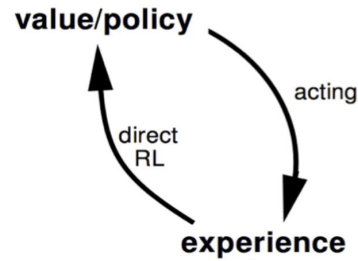


Week 6+7: Model-free reinforcement learning

Typically we don't have a model of the MDP, but only a *simulator* (or the *real-world*)

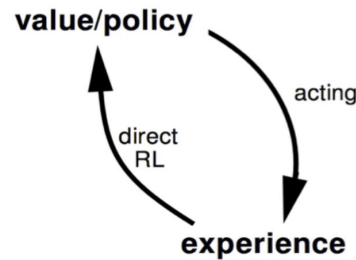
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We need to iterate:

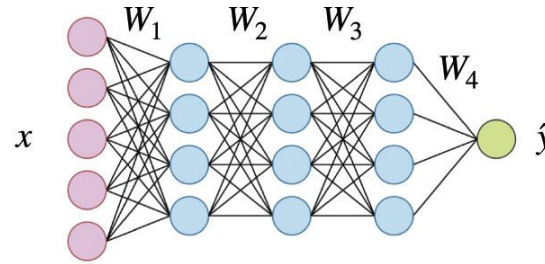
1. **Act:** sample traces (with **exploration/exploitation trade-off** – Week 2)
2. **Update:** use the obtained data to improve our solution (**credit assignment**)

Week 6+7: Model-free reinforcement learning

Key challenge = **credit assignment**

Week 6+7: Model-free reinforcement learning

Key challenge = **credit assignment**



BP
(Rumelhart et al. '86)

$$\delta W_1 \leftarrow \delta W_2 \leftarrow \delta W_3 \leftarrow e$$

$W_2^T \quad W_3^T \quad W_4^T$

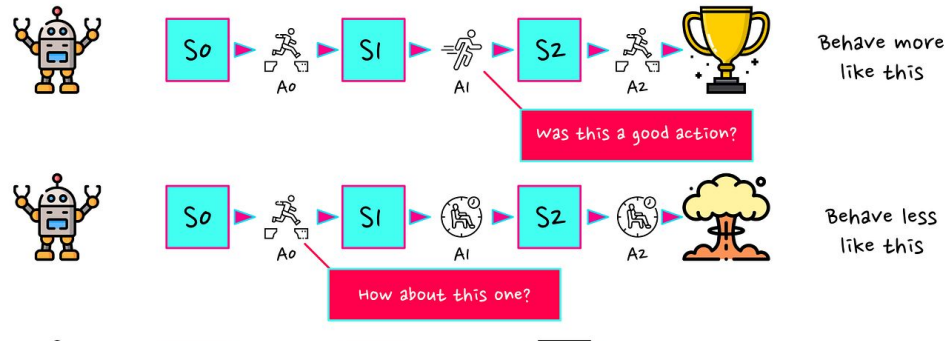
(**not** the credit assignment within a neural network – typically w. backpropagation)

Week 6+7: Model-free reinforcement learning

Key challenge = (temporal) **credit assignment**

Week 6+7: Model-free reinforcement learning

Key challenge = (temporal) **credit assignment**



(which actions in a sequence contributed to a good or bad outcome)

Week 8-9: Retake week / Free

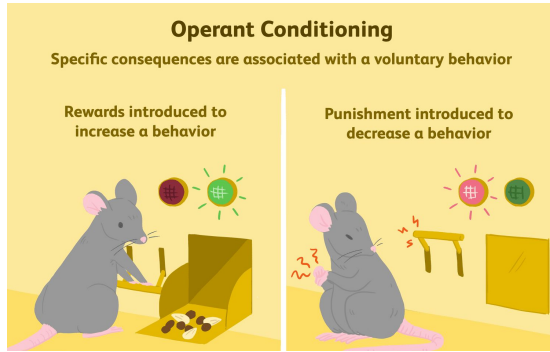
Week 8-9: Retake week / Free



Week 10: Psychology & Neuroscience

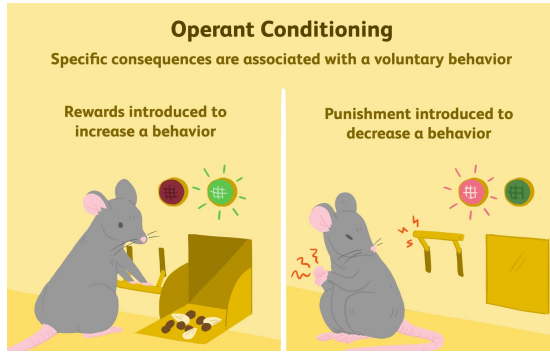


Week 10: Psychology & Neuroscience

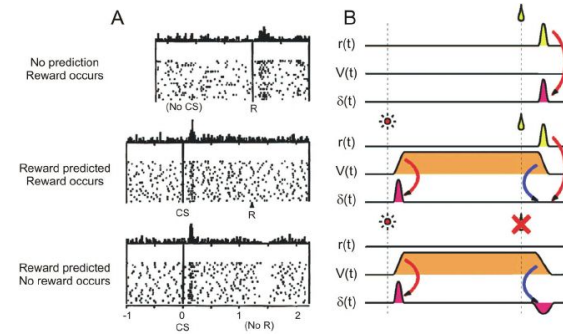


Psychology: inspiration in instrumental conditioning

Week 10: Psychology & Neuroscience



Psychology: inspiration in instrumental conditioning



Neuroscience: algorithmic RL concepts empirically match dopamine signalling

Decision-making problems

One-step + model unknown

Sequential + model known

Exploration

**Markov Decision Process
Dynamic Programming**

Q&A + Quiz + A1

Sequential +
model unknown

Model-free reinforcement learning
(Credit assignment)

Psychology & neuroscience

Q&A + Quiz + A2

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Model-free reinforcement learning

Psychology & neuroscience

(Credit assignment)

Q&A + Quiz + A2

Model unknown
but learned

Block II:

Model-based
Reinforcement Learning

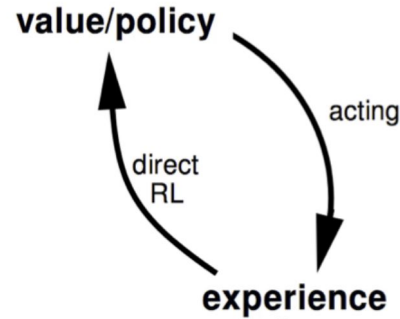
Week 12: Model-based reinforcement learning

We typically don't have a model at the start, but can we learn one from data?



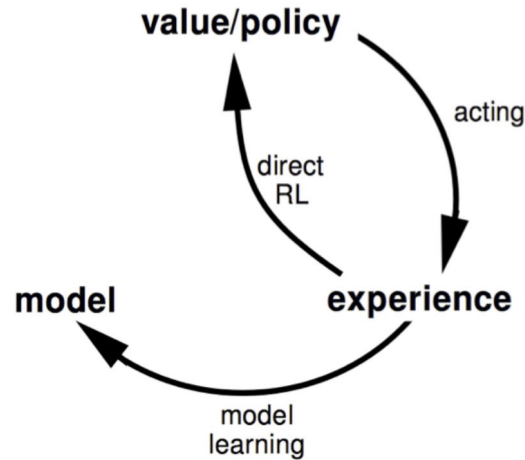
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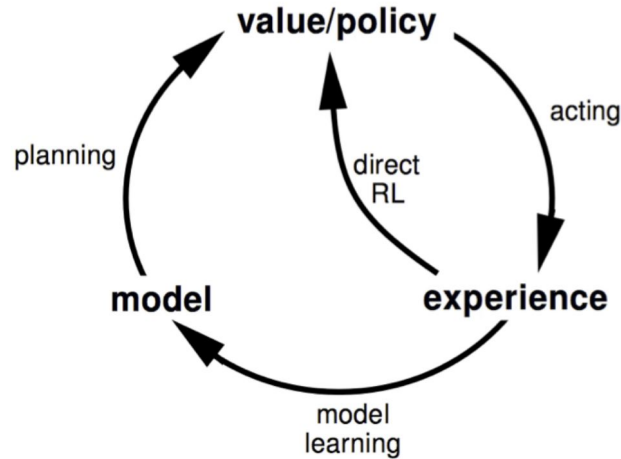
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Week 13: Decision-time planning

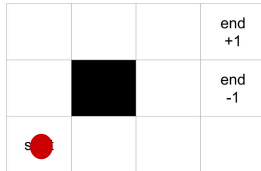


Week 13: Decision-time planning

If we have a model, we can also try to solve the problem through repeated pure planning from a given state ('decision-time planning')

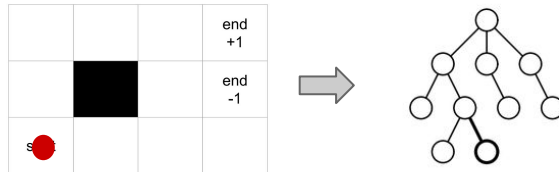
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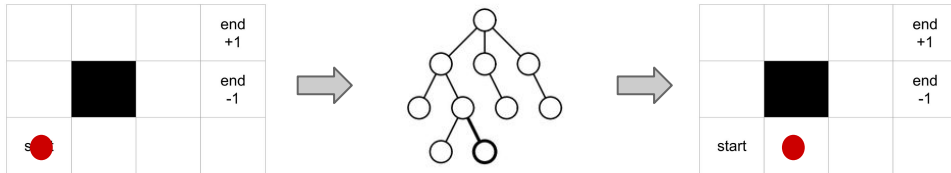
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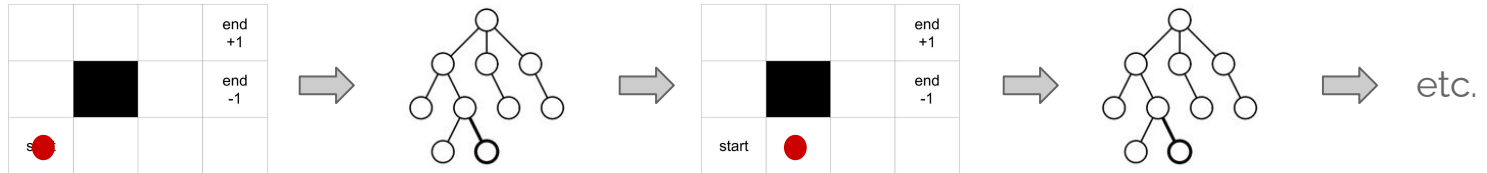
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Week 13: Sample-based planning

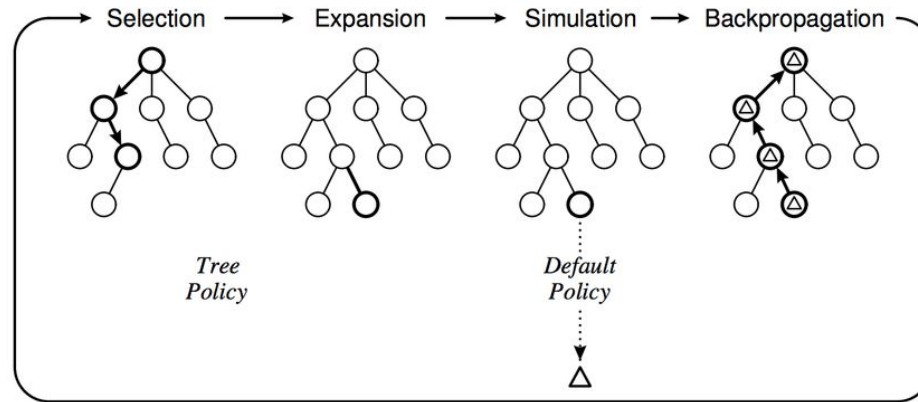


Week 13: Sample-based planning

Instead of systematic enumeration, use *statistical sampling (with uncertainty)* to expand

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Most popular class:
Monte Carlo Tree Search (MCTS)

Week 14: Advanced topics



Week 14: Approximation



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Tabular solutions don't scale to larger problems, where we need approximate solutions



Week 14: Approximation

Tabular solutions don't scale to larger problems, where we need approximate solutions

states	actions			
	a_0	a_1	a_2	$\dots$
s_0	$Q(s_0, a_0)$	$Q(s_0, a_1)$	$Q(s_0, a_2)$	$\dots$
s_1	$Q(s_1, a_0)$	$Q(s_1, a_1)$	$Q(s_1, a_2)$	$\dots$
s_2	$Q(s_2, a_0)$	$Q(s_2, a_1)$	$Q(s_2, a_2)$	$\dots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Tabular:

First part of course, focus on
RL concepts

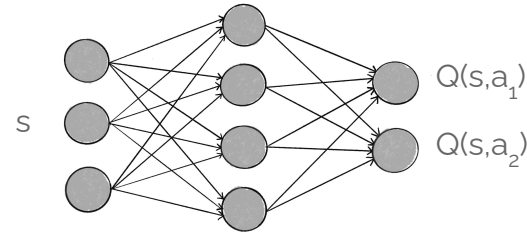
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$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Tabular:

First part of course, focus on
RL concepts



Approximate:

Deep reinforcement learning -
replace tabular solution with neural
network that generalizes

Week 14: Policy-based RL



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Instead of learning the value of actions, we may *directly learn action selection probabilities*

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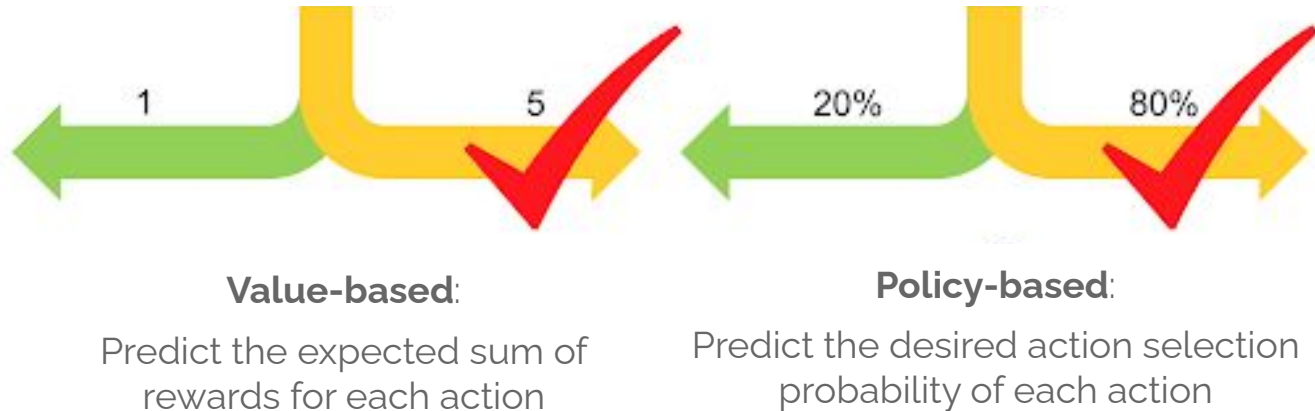


Value-based:

Predict the expected sum of
rewards for each action

Week 14: Policy-based RL

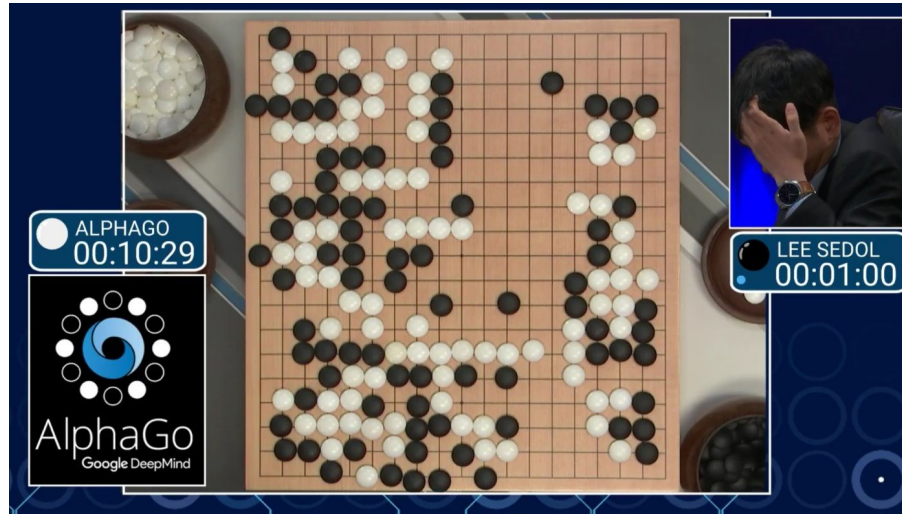
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Week 15: Applications



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AlphaGo: One of the key successes of reinforcement learning

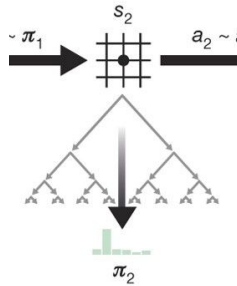
Week 15: AlphaGo

Brings together the main concepts of the whole course



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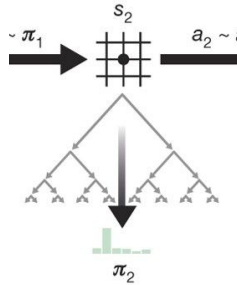
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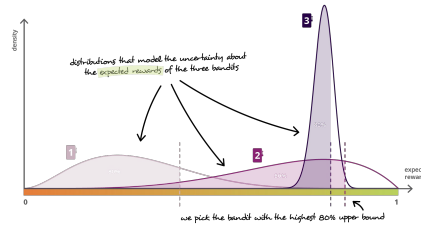
Sample-based planning
(Monte Carlo Tree Search)

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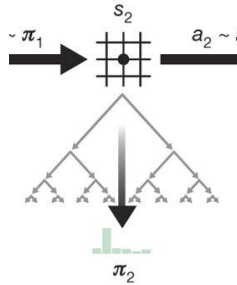
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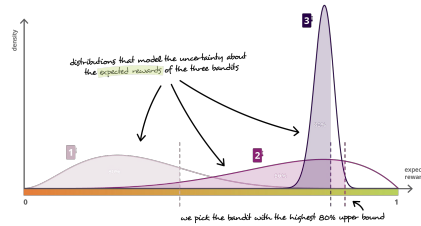
**Exploration +
Credit assignment**

Week 15: AlphaGo

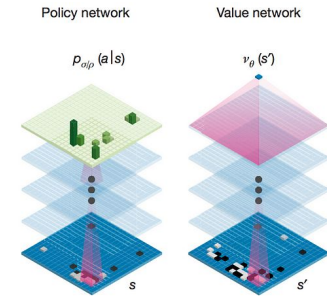
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Sample-based planning
(Monte Carlo Tree Search)



Exploration +
Credit assignment



**Neural network approximation +
policy & value (actor-critic)**

Decision-making problems

One-step + model unknown

Sequential + model known

Exploration

**Markov Decision Process
Dynamic Programming**

Q&A + Quiz + A1

Sequential +
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Model-free reinforcement learning

Psychology & neuroscience

(Credit assignment)

Q&A + Quiz + A2

Model unknown
but learned



Decision-making problems

One-step + model unknown

Sequential + model known

Exploration

**Markov Decision Process
Dynamic Programming**

Q&A + Quiz + A1

Sequential +
model unknown

Model-free reinforcement learning

Psychology & neuroscience

(Credit assignment)

Q&A + Quiz + A2

Model unknown
but learned

**Sample-based
Planning**

Model-based reinforcement learning

**Advanced topics:
Approximation & Policy-based**

Focus extra compute on current state

Deep reinforcement learning

Q&A + Quiz + A3

AlphaGo & Applications

At Home

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- Start on A1 (Colab notebooks also explain the material)

Questions?